1. First Order Linear ODEs

$y'(x) + p(x)y(x) = g(x)$.

1.1. **First Order Linear.** For each $\mu(t) \neq 0$, the ODE is equivalent to $\mu y' + \mu p y = \mu g$. Integrating factor μ satisfies $\mu' = \mu p$.

1.1.1. *Separation of Variables.* $M(x,y) + N(x,y)y' = 0$.
Separate variables; or when $\partial_2 M = \partial_1 N$ is exact, we have $M = \partial_1 \varphi$, $N = \partial_2 \varphi$, $\frac{\mathrm{d}}{\mathrm{d}t}\varphi(x,y) = 0$.

2. Second Order Linear ODEs

2.1. **Homo 2nd Order Lin, Const Coeff.** $ay'' + by' + cy = 0$, $a, b, c \in \mathbb{R}$. Ansatz: $y(t) = c_0 e^{\lambda t}$.

(i) Two real roots. $y(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$.
(ii) Two different complex roots $\lambda \pm \mathrm{i}\mu$.

$$y(t) = [c_1 \operatorname{Re} + c_2 \operatorname{Im}] e^{\lambda + \mathrm{i}\mu} = e^{\lambda t}\left[c_1 \cos \mu t + c_2 \sin \mu t\right], \quad c_1, c_2 \in \mathbb{R}.$$

(iii) One real root with a multiplicity two. Reduction of order with ansatz $y_2(t) = \mu(t) y_1(t)$. If y_1, y_2 are solutions to $Py'' + Qy' + Ry = 0$ (where P, Q, R are not necessarily constants), then μ sovles

$$\mu'' P y_1 + \mu' \left[2Py_1' + Qy_1\right] = 0 \implies \frac{\mu''}{\mu'} = -\frac{2Py_1' + Qy_1}{Py_1}.$$

2.2. **Homo 2nd Order Lin ODEs, Gen Coeff.** $Py'' + Qy' + Ry = 0$.
Ansatz $y(x) = \sum_{n \geq 0}^{\infty} a_n (x - x_0)^n$.
Theorem 2.1. *If $y'' + p(x)y' + q(x)y = 0$ is such that p, q are analytic in $B(x_0, r)$, then there exists a power series solution y to the ODE with convergence radius at least R.*

2.3. **Nonhomo 2nd Order Lin ODEs.** $Py'' + Qy' + Ry = G$.
General solution: $y = y_0 + c_1 y_1 + c_2 y_2$. Find y_0 using variation of parameters / constants, with ansatz $y_0 = \mu_1 y_1 + \mu_2 y_2$ and the restriction $\mu_1' y_1 + \mu_2' y_2 = 0$. Calculate to get $\mu_1' y_1 + \mu_2' y_2 = 0$, $\mu_1' y_1' + \mu_2' y_2' = g$.

3. First order ODE system

3.1. **1st Order Homo ODE System.** $x'(t) = A(t)x(t)$.
Definition 3.1. $x^1, \ldots, x^n$ are linear independent if the **Wronskian**, $W(t) := \det\left(x^1, \ldots, x^n\right)$, is not identically zero (equivalently, none zero anywhere, by next theorem).
Theorem 3.2. $W'(t) = \operatorname{tr}(A(t))\, W(t)$.
Theorem 3.3. *n linearly independent solutions span the solution space.*
Theorem 3.4. *Suppose that x^i solves $\frac{\mathrm{d}}{\mathrm{d}t} x^i = A(t)x^i$ with the IC $x^i(t_0) = e_i$ for $1 \leq i \leq n$. Then x^i are linearly independent solutions.*

3.2. **1st Order Lin Homo ODE System, Const Coeff.** $x'(t) = Ax(t)$.

3.2.1. *Eigenpairs.* Ansatz: $x(t) = e^{\lambda t} v$. λ, v are eigenpairs.

3.2.2. *Matrix Exponential.* Solution: $x(t) = e^{A(t - t_0)} x_0$.
$\exp[A] := \sum_{k=0}^{\infty} \frac{A^k}{k!}$.
Lemma 3.5.

(i) If $B = T^{-1}AT$, then $\exp[B] = T^{-1}\exp[A]T$.
(ii) If $AB = BA$, then $\exp[A + B] = \exp[A] \cdot \exp[B]$.
(iii) $(\exp[A])^{-1} = \exp[-A]$.
(iv) If $Av = \lambda v$, then $\exp[A]v = e^{\lambda} v$.
(v) $\frac{\mathrm{d}}{\mathrm{d}t}\exp[tA] = A\exp[tA] = \exp[tA]A$.

Remark 3.6.

$$\exp[Nt] = \begin{pmatrix} 1 & t & \frac{t^2}{2!} & \ldots & \frac{t^{l-1}}{(l-1)!} \\ 0 & 1 & t & \ldots & \frac{t^{l-2}}{(l-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix}$$

◇

3.3. **Nonhomo 1st Order Lin ODE Systems.** $x' = A(t)x + G(t)$, $x(t_0) = x_0$. Solution: $x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^{t} e^{A(t-s)} G(s)\, \mathrm{d}s$.

4. Existence and Uniqueness

Theorem 4.1 (Existence and Uniqueness)**.** *$y' = f(t, y(t))$, $y(t_0) = y_0$. In $R : |t - t_0| \leq a, |y - y_0| \leq b$, f is continuous and Lipschitz in y with Lip const L. For $0 < h < \min\{a, b/M, 1/L\}$, the ODE admits a unique C^1 solution for $|t - t_0| \leq h$.*
Remark 4.2. Continuity is sufficient for existence. Lipschitz is sufficient for uniqueness. ◇

5. Quantitative Estimates

Lemma 5.1 (Gronwall's Inequality)**.** *For given α, β, if $\beta(t) \geq 0$, $\psi(t) \leq \alpha(t) + \int_0^t \beta(s)\psi(s)\, \mathrm{d}s$, $B(s) := \exp\left(\int_0^s \beta(t)\, \mathrm{d}t\right)$, then*

$$\psi(t) \leq \alpha(t) + \int_0^t \alpha(s)\beta(s)\frac{B(t)}{B(s)}\, \mathrm{d}s = \alpha(0)B(t) + \int_0^t \alpha'(s)\frac{B(t)}{B(s)}\, \mathrm{d}s.$$

Remark 5.2. • $\alpha = 0$ implies $\psi \leq 0$.
• $\alpha(t) = \alpha_0 + \alpha_1 t$, $\beta = L$. Then $\psi(t) \leq \alpha_0 e^{Lt} + \alpha_1 \int_0^t e^{L(t-s)}\, \mathrm{d}s$.
◇

Theorem 5.3 (Continuous Dependence)**.** *Suppose f is Lipschitz in the domain D of interest with Lipschitz constant L and denote $M := \sup_D |f - g|$. Then, $|x(t) - y(t)| \leq e^{L|t-t_0|}|x_0 - y_0| + \frac{M}{L}\left(e^{L|t-t_0|} - 1\right)$.*
Theorem 5.4 (Comparison)**.** *Suppose F is Lipschitz. Let $f, g \in C^1$ be such that $f' \leq F(t, f(t))$ and $g' = F(t, g(t))$. If $f(a) \leq g(a)$, then $f(t) \leq g(t)$ for all $t \in [a, b]$.*

5.1. **Extension.**
Lemma 5.5 (Gluing)**.** *Suppose $f \in C(\mathbb{R}, \mathbb{R})$ and $f(t, x)$ is Lipschitz in any bounded region. Let φ_i be a solution of $x' = f(t, x)$, $x(t_0) = x_0$ on the interval $I_i = (a_i, b_i)$. If $t_0 \in I_1 \cap I_2 = (a, b)$ and $\varphi_1(t_0) = \varphi_2(t_0)$, then φ_1 and φ_2 agrees on $I_1 \cap I_2$ and*

$$\varphi(t) := \begin{cases} \varphi_1(t), & t \in I_1 \\ \varphi_2(t), & t \in I_2 \end{cases}$$

is a C^1 solution to the ODE on $I_1 \cup I_2$.

6. Blowup

Theorem 6.1 (Blowup Criterion)**.** *Suppose $f \in C(\mathbb{R}, \mathbb{R})$ is Lipschitz in x in any bounded region. Suppose that φ is a C^1 solution to $\frac{\mathrm{d}}{\mathrm{d}t}x = f(t, x)$ on (t_-, t_+). The solution can be extended to $(t_-, t_+ + \varepsilon)$ for some $\varepsilon > 0$ if and only if one of the following equialent conditions hold:*

(i) There exists a sequence $\{t_n\}$ such that $t_n \to (t_+)^-$ and $\varphi(t_n) \to \varphi_0 \neq \pm\infty$. Note that this is true for oscillating behavior but not monotonic blowup.
(ii) $\liminf_{t \to (t_+)^-} |\varphi(t)| < +\infty$.

In particular, the solution cannot be extended beyond t_+ if $\lim_{t \to (t_+)^-} |\varphi(t)| = +\infty$.
Proposition 6.2. *Consider $x' = f(t, x)$ with $|f| \leq M$ for each (t, x), $f \in C(\mathbb{R}, \mathbb{R})$ and Lipschitz in x in any bounded region. Then the solution exists globally and is unique.*

Example 6.3 (Riccatti ODE). Consider the Riccatti ODE $y' = y^p$, $y(0) = 1$.

- $p = 1$. The unique solution is $y(t) = e^t$.
- $p > 1$. By the comparison principle we have $y \geq e^t$ since $y(0) \geq 1$. By separation of variables, we have $y'/y^p = 1$ and then
$$y^{p-1}(t) = \frac{1}{1-(p-1)t} \to +\infty \text{ as } t \to \left(\frac{1}{p-1}\right)^-.$$
So we have blow up at $t_+ = 1/(p-1) < \infty$.

Example 6.4. $y' = y^2$, $y(0) = y_0$.

- If $y_0 > 0$, we have finite time blowup.
- If $y_0 = 0$, we have $y \equiv 0$.
- If $y_0 < 0$, we have $y = y_0/(1 - ty_0)$. Note that $1 - ty_0 > 0$ for $t > 0$. Thus $y_0 \sim -1/t$ as $t \to +\infty$.

Example 6.5. Consider $y' = y^2 + \varepsilon$ for $\varepsilon > 0$ and $y(0) = y_0$. Note that $y(t) \geq y_0 + \varepsilon t$ and so $y(t_0) > 0$ for t_0 large. Then, we have $\dot{y} > y^2$ and thus blowup. ◁

Example 6.6. Consider $y' = y^2 - \varepsilon^2$. We have $y_1 \equiv \varepsilon$ and $y_2 \equiv -\varepsilon$. For $y_0 > \varepsilon$ we again have $y' \geq y^2$ and so blowup. ◁

7. Autonomous ODE and Phase Portrait/Plane

7.1. **Phase Portrait of 2 x 2 Systems.** Consider the linear 2×2 system: $x' = Ax$. It has equilibrium at $x_* = 0$.
Case 1: $\lambda_1 < \lambda_2$ real. For large t, $x(t) \approx e^{\lambda_2 t} c_2 v_2$.
Case 2: $\lambda = \alpha \pm \mathrm{i}\beta$, $v = a \pm \mathrm{i}b$, $b \neq 0$. Solution: $x(t) = e^{\alpha t} V(t)$, where V is periodic. Stability depends on the sign of α.
Case 3: One real eigenvalue (with one or two linearly independent eigenvectors). Solution: $x(t) = e^{\lambda t}\,(c_1 v_1 + c_2(tv_1 + v_2))$.
Case 4: One eigenvector v_1 and one generalized eigenvector v_2 such that $(A - \lambda)^2 v_2 = 0$. General solution: $x(t) = e^{\lambda t} t \left(\frac{1}{t} c_1 v_1 + c_2 v_2\right)$. For large t, the trajectory is approximately along the line on v_2.

7.2. **Stability.**
Lemma 7.1. *For each $\varepsilon > 0$ and integer $n \geq 0$, there exists a constant $C > 0$ such that $t^n \leq Ce^{\varepsilon t}$ for $t \geq 0$.*
Theorem 7.2 (Linear Stability)**.** *Consider the ODE system $x' = Ax$, $A \in \mathbb{R}^{n\times n}$, with $x_* = 0$ as a critical point. Let λ_i be the eigenvalues of A.*

(i) If $\operatorname{Re}\lambda_i < 0$ for each i, then x_* is stable and asymptotically stable.
(ii) If $\operatorname{Re}(\lambda_i) > 0$ for some i, then x_* is unstable.
(iii) If $\operatorname{Re}(\lambda_i) \geq 0$ for some i, then x_* is not asymptotically stable.
(iv) If $\operatorname{Re}(\lambda_i) \leq 0$ for each i and if the Jordan block J_i associated with λ_i, $\operatorname{Re}(\lambda_i) = 0$, is 1×1, then x_* is stable. If the block is larger than 1×1, then x_* is unstable.

7.3. **Nonlinear Stability.**
Theorem 7.3 (Nonlinear Stability)**.** *Let λ_i be the eigenvalues of $A := \nabla F(x_*)$, where $F \in C^2$.*

(i) If $\operatorname{Re}(\lambda_i) < 0$ for each i, then x_* is asymptotic stable and stable.
(ii) If for some i we have $\operatorname{Re}\lambda_i > 0$, then x_* is unstable.

In other cases (say $\operatorname{Re}\lambda_i = 0$) it is unclear (we need some information from G.)

7.4. **Liapunov.** $\dot{L}(x(t)) := \frac{\mathrm{d}}{\mathrm{d}t} L(x(t)) = F(x(t)) \cdot \nabla L(x(t))$.
Theorem 7.4 (Liapunov Stability)**.** *Let x_* be an equilibrium of the autonomous system $x' = F(x)$, $F \in C^1$. Assume that there exists L such that $L(x_*) = 0$ and $L(x) > 0$ for each $x \neq x_*$.*
If $\dot{L}(x) \leq 0$ for each $x \in D \setminus \{x_*\}$, then x_* is stable. If $\dot{L}(x) < 0$ for each $x \neq x_*$, then x_* is asymptotically stable.

In the first case, for $\alpha := \min_{\partial B_\varepsilon} L$, pick δ such that $\sup_{B_\delta} L < \alpha$. Then any solution starting in $B_\delta(x_*)$ will stay in $B_\varepsilon(x_*)$.

8. Periodic Solutions, The Poincare-Bendixson Theorem

This section concerns the 2×2 autonomous system $x' = F(x)$ where F is locally Lipschitz in x.

8.1. **Limits Sets and Invariant Sets.**
Definition 8.1.

◁ (i) Use $\varphi_t(x)$ to denote the solution with IC x at time 0.
(ii) The trajectories forward and backward in time are $\gamma_+(x)$ and $\gamma_-(x)$.
(iii) The **omega limit set** and the **alpha limit set** are
$$\omega(x) := \left\{y : \varphi_{t_n}(x) \to y \text{ for some } t_n \to \infty\right\},$$
$$\alpha(x) := \left\{y : \varphi_{t_n}(x) \to y \text{ for some } t_n \to -\infty\right\}.$$
◁(iv) A set D is **positively invariant** if for each $x \in D$, $\gamma_+(x) \subset D$.

8.1.1. *Some Immediate Properties.*
Proposition 8.2. *φ_t has a semi group property: $\varphi_{t+s} = \varphi_t \circ \varphi_s$.*
Proposition 8.3. *If x, z lie on the same trajectory, then $\omega(x) = \omega(z)$ and $\alpha(x) = \alpha(z)$.*
Proposition 8.4. *Suppose D is positive invariant and closed. If $x \in D$, then $\omega(x) \subset D$. In particular, $\omega(D) \subset D$.*
Proposition 8.5. *$\omega(x)$ is a closed positive invariant set.*

8.2. **Local Sections and Flow Boxes.** Fix planar system $x' = F(x)$. The **transverse line at x_0 $\ell(x_0)$** is parameterized as $h : \mathbb{R} \to \ell(x_0)$, $h(u) := x_0 + uv_0$, where $v_0 \perp F(x_0)$ is a unit vector.
By continuity, $F(x) \cdot F(x_0) > 0$ for x in $|x - x_0| < \varepsilon$ for $\varepsilon > 0$ small.
The **local section** at x_0 is $S := \{h(u) \in \ell(x_0), |u| < \varepsilon\} \subset \ell(x_0)$.
Write $\Psi(s, u) := \varphi_s[h(u)]$. For $U_\sigma := \{|u| \text{ small}, |s| < \sigma\}$, the **flow box at x_0** is $V_\sigma := \Psi(U_\sigma$. For σ small, Ψ is 1-1. In particular, we can "pull back" $x \in V_\sigma$ to the section S by flowing backwards for time s:
$$\varphi_{-s}(x) = \varphi_{-s}(\varphi_s(h(u))) \in S.$$
Proposition 8.6. *Suppose $\varphi_{t_0}(z_0) = x_0$ (note that we do not require t_0 to be small). Then there exists an open U containing z_0 and a continuous function $\tau \in C^0(U)$ such that*
$$\varphi_{\tau(z)}(z) \in S_{x_0}, \quad \forall z \in U$$
and $\tau(z_0) = t_0$. Note that τ is interpreted as the arrival time to the section S_{x_0}.

8.3. **The Poincare Map.** The **Poincare map** $P : S_{x_0} \supset D_P \to S_{x_0}$ sends each $x \in S_{x_0}$ to its first return point on $S(x_0)$, if it does return. An application of the implicit function theorem as in the previous proposition guarantees that P is defined and C^1 in a neighborhood of x_0 in S_{x_0}.
The Poincare map is monotone in the following sense:
Proposition 8.7. *Suppose that S_{x_0} is a local section and $y_i = \varphi_{t_i}(x) \in S_{x_0}$, where $t_1 < t_2 < t_3 < \ldots$. Then, $y_1, y_2, \ldots$ are monotone along S_{x_0}. That is, y_n is between y_{n-1} and y_{n+1} on S_{x_0} for each n.*
Proposition 8.8. *Suppose $y \in \omega(x)$. The solution through y crosses any S at most once.*

8.4. **Poincare-Bendixson.**
Theorem 8.9 (Poincare-Bendixson)**.** *Suppose $\omega(x)$ is nonempty, closed, bounded, and does not contain an equilibrium point. Then, $\omega(x)$ is a closed orbit.*
Corollary 8.10. *Suppose K is closed, bounded, positive (negative) invariant. Then K contains an equilibrium or a closed orbit.*